

Exercise 1a

The full hamiltonian for the system is given by

$$\mathcal{H} = \mathcal{H}_{\text{atom}} + \mathcal{H}_{\text{int}} + \mathcal{H}_{\text{field}}$$

Let $|i\rangle\langle j| = |i\rangle \otimes \langle j_k|$ be a basis for the system with $|i\rangle$ being eigenstates of the atom and $|j_k\rangle$ representing a field with j photons in mode k . Setting $|1\rangle$ to be zero energy, the atomic hamiltonian is given by

$$\mathcal{H}_{\text{atom}} = \hbar\omega_{12}|2\rangle\langle 2| + \hbar\omega_{13}|3\rangle\langle 3|$$

The radiation field has hamiltonian

$$\mathcal{H}_{\text{field}} = \sum_{k,\lambda} \hbar\nu_{k,\lambda} \left(a_{k\lambda}^\dagger a_{k\lambda} + \frac{1}{2} \right)$$

and the interaction hamiltonian is given by

$$\mathcal{H}_{\text{int}} = -\hat{d} \cdot \hat{E} = - \sum_{\alpha, \alpha'} |\alpha\rangle\langle\alpha| \hat{d} |\alpha'\rangle\langle\alpha'| \cdot \sum_{k,\lambda} \sqrt{\frac{\hbar\nu_k}{2\varepsilon_0 V}} \left(\hat{a}_{k\lambda} \hat{\varepsilon}_{k\lambda} e^{i\vec{k}\cdot\vec{r}} + \hat{a}_{k\lambda}^\dagger \hat{\varepsilon}_{k\lambda}^* e^{-i\vec{k}\cdot\vec{r}} \right)$$

By assumption, $\langle 2|\hat{d}|1\rangle = \langle 3|\hat{d}|1\rangle = \mu\hat{z}$. Letting $g_k = \frac{\mu}{\hbar} \sqrt{\hbar\nu_k/2\varepsilon_0 V} (\hat{\varepsilon}_k \cdot \hat{z}) e^{i\vec{k}\cdot\vec{r}}$, we get

$$\mathcal{H}_{\text{int}} = -\hbar \sum_{k,\lambda} g_k (|2\rangle\langle 1| + |3\rangle\langle 1|) \hat{a}_{k\lambda} + g_k^* (|1\rangle\langle 2| + |1\rangle\langle 3|) \hat{a}_{k\lambda}^\dagger$$

To study spontaneous emission, we consider states $|2, 0\rangle, |3, 0\rangle$ (atom in excited state, no photons) and $|1, 1_k\rangle$ (atom in ground state after emitting photon in mode k). We evaluate the matrix elements as

$$\begin{aligned} \langle 2, 0|\mathcal{H}|2, 0\rangle &= \hbar\omega_{12} \\ \langle 3, 0|\mathcal{H}|3, 0\rangle &= \hbar\omega_{13} \\ \langle 1, 1_k|\mathcal{H}|1, 1_k\rangle &= \hbar\nu_k \\ \langle 2, 0|\mathcal{H}|1, 1_k\rangle &= -\hbar g_k \\ \langle 3, 0|\mathcal{H}|1, 1_k\rangle &= -\hbar g_k \end{aligned}$$

where we ignored the $\hbar\nu_k/2$ term in the field hamiltonian since it adds a constant to the total energy, irrelevant to dynamics. Thus, we get

$$\mathcal{H} = \underbrace{\hbar\omega_{12}|2\rangle\langle 2| + \hbar\omega_{13}|3\rangle\langle 3| + \sum_k \hbar\nu_k |k\rangle\langle k|}_{:=\mathcal{H}_0} - \sum_k (\hbar g_k |2\rangle\langle k| + \hbar g_k |3\rangle\langle k| + \text{h.c.})$$

where we dropped the sum over λ by absorbing it implicitly into the sum over k . Let $\mathcal{H}_1$ be the interaction part of the hamiltonian, i.e $\mathcal{H}_1 = \mathcal{H} - \mathcal{H}_0$. We move to the interaction picture w.r.t $\mathcal{H}_0$ by defining $|\psi_I\rangle = \hat{U}^{-1}|\psi\rangle$ where $\hat{U} = e^{-i\mathcal{H}_0 t/\hbar}$. Then,

$$\begin{aligned} i\hbar \frac{d|\psi_I\rangle}{dt} &= i\hbar \frac{d\hat{U}^{-1}}{dt} |\psi\rangle + i\hbar \hat{U}^{-1} \frac{d|\psi\rangle}{dt} \\ &= -\mathcal{H}_0 \hat{U}^{-1} |\psi\rangle + \hat{U}^{-1} (\mathcal{H}_0 + \mathcal{H}_1) |\psi\rangle \\ &= \hat{U}^{-1} \mathcal{H}_1 |\psi\rangle \\ &= \hat{U}^{-1} \mathcal{H}_1 \hat{U} |\psi_I\rangle \end{aligned}$$

where we used the fact that $\hat{U}^{-1}$ commutes with $\mathcal{H}_0$ in the last step. Thus, our effective hamiltonian in the interaction picture is $\mathcal{H}_{\text{eff}} = \hat{U}^{-1} \mathcal{H}_1 \hat{U}$. We note that since $\mathcal{H}_0$ is diagonal, we have

$$\hat{U}^{-1} = e^{i\mathcal{H}_0 t/\hbar} = e^{i\omega_{12}t} |2\rangle\langle 2| + e^{i\omega_{13}t} |3\rangle\langle 3| + \sum_k e^{i\nu_k t} |k\rangle\langle k|$$

Hence, we get

$$\mathcal{H}_{\text{eff}} = -\hbar \sum_k g_k |2\rangle\langle k| e^{-i(\nu_k - \omega_{12})t} + g_k |3\rangle\langle k| e^{-i(\nu_k - \omega_{13})t} + \text{h.c.}$$

Exercise 1b

Apply the unitary operator $\hat{U} = e^{-i\sum_k (\nu_k - \omega_{12})|k\rangle\langle k|}$. We write $|\psi_I(t)\rangle = c_2(t)|2\rangle + c_3(t)|3\rangle + \sum_k c_k(t)|k\rangle$. Using the effective hamiltonian from part (a), we find

$$\begin{aligned} i\hbar \frac{d}{dt} |\psi_I(t)\rangle &= \mathcal{H}_{\text{eff}} |\psi_I\rangle \\ &= -\hbar \sum_k (g_k^* c_2(t) e^{i(\nu_k - \omega_{12})t} + g_k^* c_3(t) e^{i(\nu_k - \omega_{13})t}) |k\rangle \\ &\quad - \hbar \sum_k g_k c_k(t) e^{-i(\nu_k - \omega_{12})t} |2\rangle + g_k c_k(t) e^{-i(\nu_k - \omega_{13})t} |3\rangle \end{aligned}$$

We equate this to $i\hbar(\dot{c}_2(t)|2\rangle + \dot{c}_3(t)|3\rangle + \sum_k \dot{c}_k(t)|k\rangle)$ and use the orthogonality of the basis kets to get the system

$$\begin{aligned} \dot{c}_2(t) &= i \sum_k g_k c_k(t) e^{-i(\nu_k - \omega_{12})t} \\ \dot{c}_3(t) &= i \sum_k g_k c_k(t) e^{-i(\nu_k - \omega_{13})t} \\ \dot{c}_k(t) &= i g_k^* c_2(t) e^{i(\nu_k - \omega_{12})t} + i g_k^* c_3(t) e^{i(\nu_k - \omega_{13})t} \end{aligned}$$